Noise Models for Superconducting Qubits

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Abstract

We describe the three canonical noise contributions to the power spectral density $S(\omega)$ of a superconducting qubit: $1/f^\alpha$ noise arising from a distribution of two-level fluctuators, a white noise floor from Johnson-Nyquist thermal fluctuations, and Lorentzian peaks from individual two-level system (TLS) defects. Each model is derived from its physical origin, expressed mathematically, and discussed in the context of the filter function formalism developed in documents 02 and 03.

1 Overview

The noise power spectral density $S(\omega)$ of a superconducting qubit encodes the full frequency-resolved noise environment experienced by the qubit. In practice, $S(\omega)$ for a superconducting qubit is well described by the sum of three physically distinct contributions:

$$\boxed{S(\omega) = \frac{A}{\omega^\alpha} + S_0 + \sum_k \frac{A_k \gamma_k^2}{(\omega - \omega_k)^2 + \gamma_k^2}} \tag{1}$$

where the first term is $1/f^\alpha$ noise, the second is a white noise floor, and the sum represents contributions from individual TLS defects. Each term dominates in a different frequency regime: $1/f^\alpha$ at low frequencies, white noise at high frequencies, and TLS peaks at specific intermediate frequencies.

2 $1/f^\alpha$ Noise

2.1 Physical Origin

The qubit frequency ω_0 is sensitive to environmental parameters — charge on the island, magnetic flux through the loop, critical current of the Josephson junction. Each of these is perturbed by fluctuators: charge traps, magnetic impurities, and atomic tunneling systems in the substrate and oxide layers. Each fluctuator switches randomly between two states at a characteristic rate γ_k, producing a Lorentzian noise contribution:

$$S_k(\omega) \propto \frac{\gamma_k}{\omega^2 + \gamma_k^2} \tag{2}$$

When many fluctuators are present with a broad distribution of switching rates, the total noise is obtained by summing over all fluctuators. If the switching rates are distributed uniformly on a logarithmic scale — i.e., there are roughly equal numbers of slow and fast fluctuators — the distribution of rates is $P(\gamma) \propto 1/\gamma$, and the total noise becomes:

$$S(\omega) = \int_0^\infty S_k(\omega) P(\gamma)\, d\gamma \propto \int_0^\infty \frac{1}{\omega^2 + \gamma^2}\, d\gamma = \frac{\pi}{2\omega} \tag{3}$$

This integral evaluates to $\pi/2\omega$, giving $S(\omega) \propto 1/\omega$. The $1/f$ spectrum is therefore not a property of any single fluctuator — it is an emergent property of having many fluctuators with a broad distribution of switching rates [1].

2.2 Mathematical Model

The generalized $1/f^\alpha$ noise model is:

$$\boxed{S_{1/f}(\omega) = \frac{A}{\omega^\alpha}} \tag{4}$$

where A is the noise amplitude and α is the spectral exponent. For pure $1/f$ noise $\alpha = 1$. In superconducting qubits, α typically lies in the range $0.7 \lesssim \alpha \lesssim 1.3$ depending on the device and dominant noise source [2, 3].

This model diverges at $\omega = 0$, reflecting the fact that there is always a low-frequency cutoff set by the finite duration of the experiment. In numerical implementation, the value at $\omega = 0$ is set to zero.

2.3 Implications for Filter Functions

The Ramsey filter function $F_{\text{Ramsey}}(\omega, t) = 4\sin^2(\omega t/2)/\omega^2$ peaks at $\omega = 0$ and scales as t^2 there. Combined with $S_{1/f}(\omega) \propto 1/\omega$, the integrand $S(\omega)F(\omega, t)$ is heavily weighted toward low frequencies. This is why Ramsey dephasing in superconducting qubits is dominated by $1/f$ noise and why Ramsey T_2^* times are typically much shorter than echo T_2 times [3].

3 White Noise Floor

3.1 Physical Origin

At frequencies above the $1/f$ knee, the fluctuator-dominated noise falls below a constant background arising from Johnson-Nyquist thermal noise. Any resistive element at finite temperature T generates voltage fluctuations with a flat power spectral density:

$$S_V(\omega) = 4k_B T R \tag{5}$$

independent of frequency [4]. In a superconducting qubit experiment, this noise enters through control lines, amplifiers, bias resistors, and readout electronics — all of which are at temperatures above the qubit's millikelvin operating point. The result is a frequency-independent noise floor that sets the minimum achievable $S(\omega)$.

3.2 Mathematical Model

$$\boxed{S_{\text{white}}(\omega) = S_0} \tag{6}$$

where S_0 is a constant. The crossover frequency ω_c at which $1/f$ noise and white noise contribute equally is found by setting $A/\omega_c^\alpha = S_0$, giving $\omega_c = (A/S_0)^{1/\alpha}$.

3.3 Implications for Filter Functions

Echo sequences suppress low-frequency $1/f$ noise by refocusing slow phase accumulation. However, white noise cannot be refocused — it contributes uniformly regardless of the pulse sequence. This sets a fundamental limit on coherence time achievable through dynamical decoupling: even with infinitely many pulses, the white noise floor remains and causes irreversible dephasing [7].

4 Lorentzian TLS Peaks

4.1 Physical Origin

Individual two-level system defects — atomic tunneling systems in amorphous oxides, dangling bonds at interfaces, or trapped charges — each have a well-defined resonance frequency ω_k and a relaxation rate

γ_k [5, 6]. When a TLS couples to the qubit, it generates fluctuations at its resonance frequency, producing a peaked noise contribution. The lineshape is Lorentzian because the TLS is a damped two-state system:

$$S_{TLS,k}(\omega) = \frac{A_k \gamma_k^2}{(\omega - \omega_k)^2 + \gamma_k^2} \tag{7}$$

A slowly relaxing TLS (γ_k small) produces a sharp, narrow peak. A fast-relaxing TLS produces a broad, low peak. In real devices, multiple TLS defects contribute at different frequencies, producing a forest of Lorentzian peaks superimposed on the $1/f$ + white background [5].

4.2 Mathematical Model

$$\boxed{S_{TLS}(\omega) = \frac{A\gamma^2}{(\omega - \omega_0)^2 + \gamma^2}} \tag{8}$$

where A is the peak amplitude, ω_0 is the TLS resonance frequency, and γ is the half-width at half-maximum (HWHM), equal to the TLS relaxation rate.

4.3 Implications for Filter Functions and Resonance Condition

When a dynamical decoupling sequence's filter peak coincides with a TLS resonance — i.e., when $\omega_{\text{peak}}(n, t) = \omega_k$ — the coherence decay is dramatically enhanced rather than suppressed. This is the **resonance condition**:

$$\frac{n\pi}{t} = \omega_k \tag{9}$$

For CPMG-n with varying t, the filter peak sweeps through the spectrum and periodically resonates with TLS defects, producing characteristic dips in the coherence as a function of t. This effect has been experimentally observed in superconducting flux qubits [3] and is one of the primary signatures used to identify and locate TLS defects in device characterization.

Crucially, this same resonance condition is what makes high-n CPMG sequences useful for TLS spectroscopy: by sweeping n and t, one can map out the locations of TLS resonances in $S(\omega)$ with high frequency resolution.

5 Combined Noise Model

The full noise model used in this project is:

$$S(\omega) = \frac{A_{1/f}}{\omega^\alpha} + S_0 + \sum_k \frac{A_k \gamma_k^2}{(\omega - \omega_k)^2 + \gamma_k^2} \tag{10}$$

This model is implemented in `simulation/noise_models.py` and used as the ground-truth $S(\omega)$ for simulation validation of the inversion pipeline. The inversion problem — recovering $S(\omega)$ from measured $\chi(t)$ — is treated in documents 05–08.

References

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